

Numerical Methods

Locating Roots: Change of Sign

Change of Sign Method: If $f(x)$ is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs, then there is at least one root α in the interval (a, b) where $f(\alpha) = 0$.

Limitations of the change of sign method:

- Does not work if the curve just touches the x-axis (repeated root)
- May miss multiple roots in the interval if there is an even number of roots
- Requires continuity -- does not work across discontinuities (e.g. $f(x) = \frac{1}{x}$)

Worked Example: Locating a Root

Show that $f(x) = x^3 - 3x + 1$ has a root in the interval $[1, 2]$.

Solution:

$$f(1) = 1 - 3 + 1 = -1 < 0$$

$$f(2) = 8 - 6 + 1 = 3 > 0$$

Since $f(1) < 0$ and $f(2) > 0$ and f is continuous, there is a root in $(1, 2)$ by the change of sign principle.

Exam Tip: When asked to "show that a root lies in an interval", you must: (1) evaluate $f(x)$ at both endpoints, (2) state the signs, (3) state that f is continuous, (4) conclude there is a change of sign so a root exists. All four steps are needed for full marks.

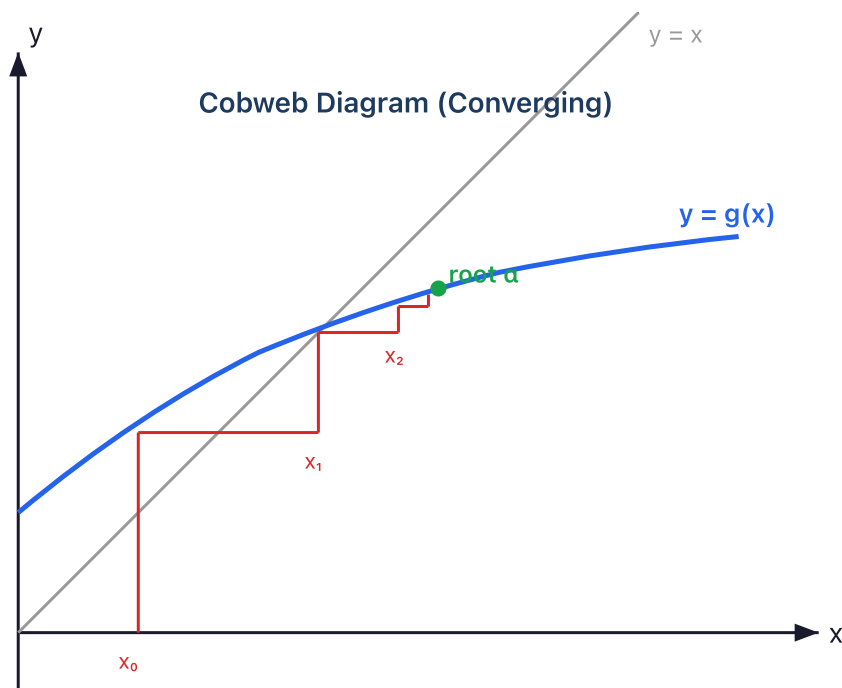
Fixed-Point Iteration: $x_{n+1} = g(x_n)$

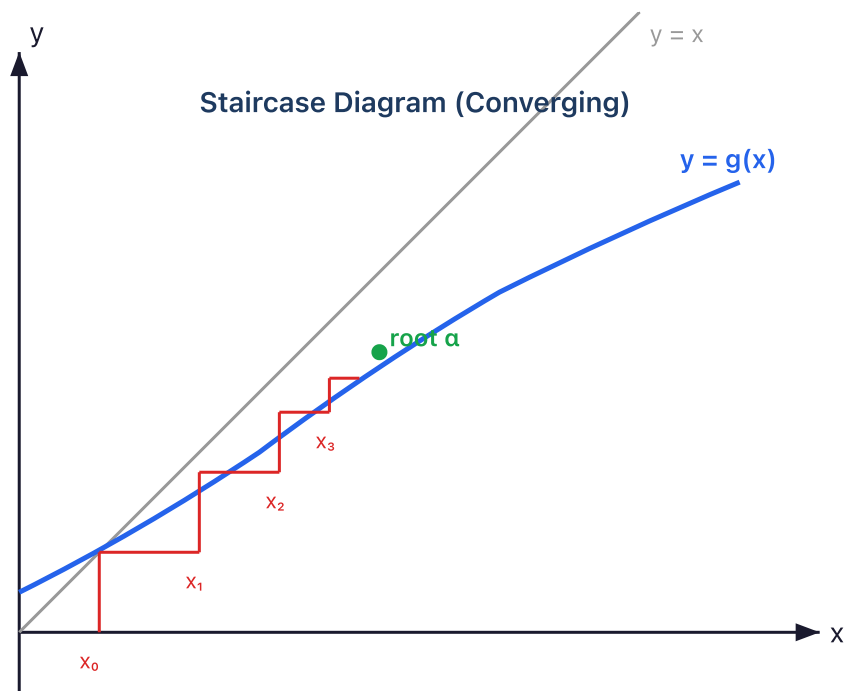
Fixed-Point Iteration: Rearrange $f(x) = 0$ into the form $x = g(x)$. Starting from an initial estimate x_0 , generate the sequence:

$$x_{n+1} = g(x_n)$$

If the sequence converges, it converges to a root α where $\alpha = g(\alpha)$.

The iteration converges when $|g'(\alpha)| < 1$ at the root. It diverges when $|g'(\alpha)| > 1$.





Worked Example: Fixed-Point Iteration

The equation $x^3 - 3x + 1 = 0$ can be rearranged to $x = \frac{x^3 + 1}{3}$. Use the iteration $x_{n+1} = \frac{x_n^3 + 1}{3}$ with $x_0 = 0.5$ to find a root to 3 decimal places.

Solution:

n	x_n
0	0.5
1	0.375
2	0.350911...
3	0.347736...
4	0.347349...
5	0.347302...

6

0.347297...

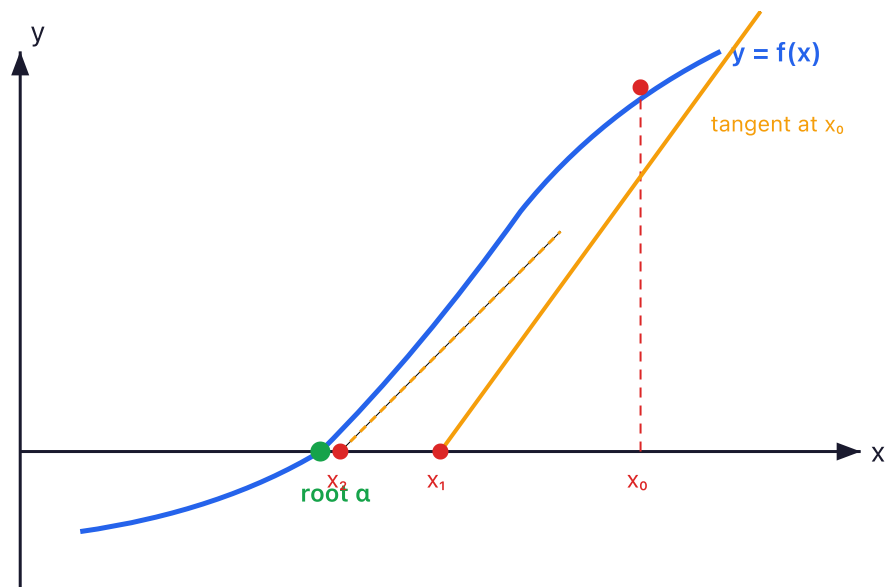
The root is $\alpha \approx 0.347$ (3 d.p.).

Newton-Raphson Method

Newton-Raphson Formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

This uses the tangent at $(x_n, f(x_n))$ to find a better approximation to the root.



Worked Example: Newton-Raphson

Use Newton-Raphson with $x_0 = 2$ to find a root of $f(x) = x^3 - 2x - 5$.

Solution: $f'(x) = 3x^2 - 2$.

$$x_1 = 2 - \frac{f(2)}{f'(2)} = 2 - \frac{8 - 4 - 5}{12 - 2} = 2 - \frac{-1}{10} = 2.1$$

$$x_2 = 2.1 - \frac{(2.1)^3 - 2(2.1) - 5}{3(2.1)^2 - 2} = 2.1 - \frac{0.061}{11.23} = 2.0946...$$

The root is approximately $x \approx 2.095$ (3 d.p.).

Exam Tip: Newton-Raphson usually converges much faster than fixed-point iteration, but can fail if $f'(x_n) = 0$ or if the initial estimate is too far from the root. Always check that your sequence is converging.

Convergence and Divergence

For fixed-point iteration $x_{n+1} = g(x_n)$:

Condition at Root α	Behaviour	Diagram Type
$ g'(\alpha) < 1$	Converges to α	Cobweb or staircase spiralling inward
$ g'(\alpha) > 1$	Diverges from α	Cobweb or staircase spiralling outward
$0 < g'(\alpha) < 1$	Converges monotonically	Staircase converging
$-1 < g'(\alpha) < 0$	Converges with oscillation	Cobweb converging

Exam Tip: If an iteration diverges, try a different rearrangement of $f(x) = 0$ into $x = g(x)$. The same equation can have multiple rearrangements, some converging and some diverging. Check that $|g'(x)| < 1$ near the root.

